

EDUARDO ADAME SALLES

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About me

I am currently a Master's student in Applied Mathematics and Data Science at Fundação Getulio Vargas (FGV EMap). I hold a Bachelor's degree in Data Science and Artificial Intelligence, graduated with honors and awarded the Undergraduate Research Distinction—granted to only one student per graduating class.

During my technical training in Industrial Mechanics, I earned distinctions in national math olympiads, notably OBMEP—the world's largest, with over 18 million students. These achievements deepened my passion for mathematics and led to a full scholarship through the CDMC Talent Program.

At FGV, I have embraced teaching opportunities, accumulating four years of experience as a teaching assistant. I also joined the Scientific Initiation and Master's Track Program early in my undergraduate studies, completing all master's-level coursework by graduation, with only the thesis remaining.

My current research focuses on exact Markov Chain Monte Carlo methods for the Normalized Power Prior, under the supervision of Prof. Luiz Max de Carvalho (FGV EMap) and co-supervision of Prof. Flávio Bambirra (UFMG). I also have been working with uncertainty quantification for black-box generative models.

From February 2023 to June 2025, I worked as a consultant for The World Bank, where I supported data collection and pre-processing, as well as the development and refinement of econometric models using PyTorch.

Educational Background

› MS. in Applied Mathematics and Data Science

Mar/2025 – Mar/2027

School of Applied Mathematics, Fundação Getulio Vargas (FGV EMap).
Rio de Janeiro, Brazil.

Full scholarship estimated in 220k BRL ($\approx$ 40k USD by May 2025).

› BS. in Data Science and Artificial Intelligence - GPA: 9.31/10

Feb/2021 – Dec/2024

School of Applied Mathematics, Fundação Getulio Vargas (FGV EMap).
Rio de Janeiro, Brazil.

Graduated with honors and received the Undergraduate Research Distinction—only one student per graduating class.

Graduate-level (MS. and PhD.) courses taken with grade A during undergraduate studies: Advanced Linear Algebra, Numerical Analysis, Visual Data Analysis, Computational Statistics, Statistical Inference, Probability Theory, Real Analysis, Mathematical Biology, Stochastic Processes, Stochastic Calculus and Causal Inference.

Full scholarship from FGV's Center for the Mathematics and Science Development (FGV CDMC), achieved by previous awards (Math Olympiads) during high school. Estimated in 300k BRL ($\approx$ 55k USD by May 2025).

Presentations on the 43rd National Congress of Applied and Computational Mathematics (CNMAC) and 7th Latin American Congress on Bayesian Statistics (COBAL). Awarded with the ICIAM Fellowship CNMAC 2024.

Workshop paper published on Statistical Frontiers in LLMs and Foundation Models at NeurIPS 2024.

› Technical Degree in Industrial Mechanics - GPA: 9.6/10

2018 – 2020

Federal Center for Technological Education Celso Suckow da Fonseca (CEFET/RJ).
Angra dos Reis, Rio de Janeiro, Brazil.

Professional Experience

> Consultant

Feb/2023 – Jun/2025

The World Bank.

I worked in developing, applying, fitting, improving, and optimizing econometric models for geospatial land use data in Latin America and the Caribbean (LAC). This project was supervised by Professors Marcelo Sant'Anna (FGV EPGE), Rafael Araújo (FGV EESP), and Francisco Costa (University of Delaware). I handled an encrypted dataset comprising tens of millions of entries, for which I was also responsible for data cleaning and processing, focusing on scalability with PyTorch and Nvidia CUDA. This assignment resulted in a submission for the Nature Sustainability journal.

Academic Experience

> Teaching Assistant

Feb/2022 – Today

School of Applied Mathematics at FGV.

Courses:

- » Autumn 2022 - Introduction to Computing. Instructor: Prof. Rafael de Pinho, PhD.
- » Spring 2022 - Multivariate Calculus. Instructor: Prof. Maria Izabel Camacho, PhD.
- » Summer 2023 - Multivariate Calculus. Instructor: Prof. Sônia Durães, PhD.
- » Autumn 2023 - Data Science Techniques and Algorithms. Instructor: Prof. Alberto Paccanaro, PhD.
- » Spring 2023 - Statistical Inference. Instructor: Prof. Philip Thompson, PhD.
- » Summer 2024 - Multivariate Calculus. Instructor: Prof. Sônia Durães, PhD.
- » Summer 2024 - Introduction to Real Analysis (Graduate-level). Instructor: Prof. Philip Thompson, PhD.
- » Autumn 2024 - Statistical Modelling. Instructor: Prof. Luiz Max Carvalho, PhD.
- » Spring 2024 - Numerical Analysis (Graduate-level). Instructor: Prof. Hugo de la Cruz Cancino, PhD.
- » Summer 2025 - Multivariate Calculus. Instructor: Prof. Sônia Durães, PhD.
- » Autumn 2025 - Introduction to Computing. Instructor: Prof. Flávio Codeço Coelho, PhD.

> Vice President

Aug/2021 – Aug/2022

Applied Mathematics Academic Board at FGV EMap (DAMA).

Selected by the student body of FGV EMap. Among my achievements during this period, the highlight is the creation and execution of the first SEEMAp – Week of Extension of the School of Applied Mathematics – which has since become an annual event, gathering over 200 students and teachers from across Brazil.

> Teaching Assistant

Feb/2019 – Jul/2020

Federal Center for Technological Education (CEFET/RJ).

Teaching Assistant during the “Fundamentals of Mechanical Design” course (Autumn 2019, Spring 2019 and Autumn 2020).

Awards

🏆 Scientific Initiation scholarship. Project: “Shape-constrained Gaussian Processes” - National Institute of Mathematics Science and Technology (INCTMat), 2023.

🏆 Silver Medal - Elon Lages Competition, Brazilian Mathematical Olympiad (OBM), 2022.

🏆 Gold Medal (1st Place) - Science, Technology and Innovation Fair of the State of Rio de Janeiro (FECTI), 2021.

🏆 Silver Medal - Brazilian Mathematical Olympiad of Public Schools (OBMEP), 2019.

🏆 Bronze Medal - Brazilian Mathematical Olympiad of Public Schools (OBMEP), 2018.

Activities

> Presentations on Events

2024 **Bayesian Shape-constrained Curve-fitting with Gaussian Processes.**

Presented in the 43rd National Congress of Applied and Computational Mathematics (CNMAC) and in the 7th Latin American Meeting on Bayesian Statistics (COBAL). Curve fitting is essential for making predictions and understanding relationships between variables across various fields, including finance, biology, and engineering. Our framework integrates prior information about the function's shape, such as monotonicity and convexity, into the modeling process to enhance the accuracy of function approximation under constraints. This approach is particularly beneficial in scenarios with limited data due to budget restrictions.

2022 **Applications of Machine Learning in Engineering.**

Presented at the Week of Teaching, Research, and Extension (SEPEX) as a guest of the director and faculty of CEFET/RJ UnED Angra dos Reis, my alma mater in Mechanics. The presentation aimed to demystify the initial steps for students in Mechanical, Electrical, and Metallurgical Engineering in Machine Learning techniques. The approach was not to delve into mathematical details but to explain in a simple, visual, and intuitive manner how to apply a model in a project in these fields.

2021 **Drone Application in Firefighter Support.**

A project I co-developed with two colleagues from the Stardust team and the Technical Course in Mechanics, guided by Professors Marcelo dos Reis Faria, PhD, and Paulo Victor dos Santos, MSc. The project gained significant recognition, winning first place at the CEFET/RJ SEPEX for all UnEDs and at the Rio de Janeiro State Science, Technology, and Innovation Fair (FECTI). We also became finalists at the Brazilian Science and Engineering Fair (FEBRACE).

> Participation on Congresses

2024 **7th Latin American Congress on Bayesian Statistics (COBAL)**

2024 **43rd National Congress of Applied and Computational Mathematics (CNMAC)**

2024 **12th Bachelier World Congress of the Bachelier Finance Society**

2024 **14th Brazilian Workshop on Continuous Optimization (BrazOpt)**

2024 **Tropical Probabilistic AI School (Prob.AI)**

2023 **42nd National Congress of Applied and Computational Mathematics (CNMAC)**

2023 **1st Latin American Congress on Industrial and Applied Mathematics (LACIAM)**

> Work in Progress

2024 – Today **Image super-resolution with guarantees via conformalized generative models**
joint work with Daniel Csillag (FGV EMap) and Guilherme T. Goedert (FGV EMap).

2024 – Today **Exact Markov chains Monte Carlo methods for the normalized power prior**
joint work with Luiz Max Carvalho (FGV EMap) and Flávio Bambirra (UFMG)

2023 – Today **Efficient function emulation for the normalized power prior**
joint work with Luiz Max Carvalho (FGV EMap), Andrew Manderson (MRC Biostatistics) and Joseph G. Ibrahim (University of North Carolina)

› Research Projects

Jul/2023 – Oct/2024

Shape-constrained Gaussian Processes

Supervised by Professor Luiz Max de Carvalho, PhD.

The use of Gaussian processes is a powerful technique for estimating functions which cannot be evaluated across their entire domain, allowing for the quantification of estimation uncertainties, especially when following the Bayesian paradigm. However, when we have prior knowledge about the shape properties of a function, especially its derivatives, it is possible to enhance the quality of our estimates. This project focuses on the development and application of shape-constrained Gaussian processes to improve function estimation. The project repository is available on: <https://github.com/adamesalles/shape-constrained-gaussian-processes>. This project became the base for my undergraduate thesis.

Feb/2018 – Dec/2021

STARDUST Drone Team

As the captain from 2019 to 2021, I led this team focused on the study, development, customization, and application of Unmanned Aerial Vehicles (UAVs). The team, comprising both faculty and students at technical and engineering levels, achieved significant milestones. In 2019, we secured the eighth position in the Formula Drone SAE Brasil competition. In 2021, we ranked first in both the CEFET/RJ extension week (across all units) and the XV FECTI by the Fundação CECIERJ. Based at CEFET/RJ UnED Angra dos Reis, our team has gained notable recognition in the state, leading to an invitation to the Brazilian Science and Engineering Fair where we became finalists in 2022.

› Courses and Materials Developed

2024

Web Development Course - FGV

Designed and produced a complete HTML and CSS course—comprising written materials, slide decks, and recorded video lectures—for Fundação Getulio Vargas. Initially developed as an asynchronous online module for undergraduate students, the course is now also offered in an *in-company* format. Available at <https://educacao-executiva.fgv.br/cursos/online/curta-media-duracao-online/desenvolvimento-de-sistemas-web-css-e-html>.

2023

L^AT_EX Course - FGV

Developed a comprehensive L^AT_EX course, including text, presentations, and on-demand videos, for the Fundação Getulio Vargas. This course is offered as an asynchronous online module for undergraduate students. It is now available as an *in-company* course at: <https://educacao-executiva.fgv.br/cursos/online/curta-media-duracao-online/latex>.

2021

SymPy Course

Created a course consisting of text, a website, and videos during my technical course internship. This was designed to assist in laboratory studies and was made available through CEFET/RJ's networks. Available on: <https://github.com/adamesalles/cursosympy>.

2020

L^AT_EX Course (old)

Produced a L^AT_EX course in the form of text, presentations, and videos for my participation in events, with the content also available on YouTube: https://www.youtube.com/playlist?list=PLtp0NMJtiOH_VR4HhOQEJdVNwb2lFzqmL.

> Selected Projects

2024 **Open Labeller for Iterative Machine learning**

Joint work with Juliano Gennari (FGV EMap) and Guilherme T. Goedert (FGV EMap).

OLIM is a user-friendly labeling interface designed for individuals without prior data science knowledge. Its primary use case is to assist in labeling patient-related data extracted from medical texts. Future versions aim to support a wider range of data formats and domains. Available on <https://nanogennari.gitlab.io/olim/>.

2023 **A Bayesian approach to understanding the Homicide Rate in the City of Rio de Janeiro by administrative regions through their Social Progress Index indicators.**

This study aims to investigate the relationship between the homicide rate in the city of Rio de Janeiro and the indicators of the Social Progress Index. Our approach involves employing Bayesian methodology to estimate the parameters of three multilevel models and subsequently comparing their performance. The Social Progress Index serves as a measure of the overall quality of life and social well-being of the population, and it has been regularly published by the Pereira Passos Institute for the City of Rio de Janeiro biennially since 2016. Given the well-known issue of violence in Rio de Janeiro, the homicide rate serves as a pertinent indicator of this problem. The city is divided into 33 administrative regions, and we utilize the corresponding data throughout this research. Available on <https://github.com/adamesalles/homicide-rate-rj>.

2023 **Gaussian Processes Visual Tool**

This paper presents the Gaussian Processes Visual Tool, an interactive tool for visualizing Gaussian Processes (GPs). The tool combines the power of Svelte, D3.js, Flask, and GPyTorch to provide a flexible and user-friendly interface. With it, users can visualize and understand GPs, make observations, and explore various kernel functions. The tool offers simulation options, allowing users to set axis parameters, likelihood variance, and even upload custom datasets. It stands out for its aesthetic design, comprehensive functionalities, and the integration of multiple important technologies. It also fills a gap by offering a beautiful and useful tool that combines all essential GP functions in one place. The tool is publicly available on <https://github.com/adamesalles/gp-visual-tool/>, enabling researchers and practitioners to employ it for GP visualization, experimentation, and education.

2023 **Voyager's Journey: an interactive visualization of NASA's mission through the solar system**

Joint work with Juan Belieni (FGV EMap) and Marcelo Amaral (FGV EMap).

This article presents an interactive visualization using web technologies of the Voyager program by showing, for each probe, a simulation of the Solar System, containing the probe's position for each day, the main events of the mission inside a timeline and a gallery of images captured by them. The visualization is controllable by a player, which offers the option to play, pause, change the speed and reset the whole visualization. Available on <https://github.com/fgv-vis-2023/final-project-voyagers-journey/>.

> References

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